

ML MODEL TRAINING GUIDE & DATASET SPECIFICATION

Comprehensive Hand-off Briefing & AI-Assisted Modeling Context for Member 2

Project Title: Predictive Flood Alert System for Bengaluru (500 m Grid, 24–48h Forecast)

Course / Institutional Context: Community Project (1BCP308), DSCE Bengaluru

Prepared by: Member 1 (Geospatial Data Pipeline)

Target Audience / Recipient: Member 2 (ML Model Developer & AI Pairing Assistant)

Dataset Version: v1.0-prod (Conforms to Flood Model Dataset Specification v1.0)

Date of Delivery: October 9, 2026

1. Executive Summary & Fast-Start Guide

Welcome Member 2! This document is your comprehensive operational guide and LLM prompt context for training the machine learning models (XGBoost and sequential LSTM) for the Bengaluru Flood Prediction System. The data engineering phase is 100% complete. You do not need to download, clean, interpolate, or re-grid any raw files.

Key Quick-Start Facts:

- Primary ML Table: delivery/flood_dataset.parquet (18 MB, 1,764,158 rows × 42 columns).
- Observations: Each row represents one 500 m grid cell on one calendar day across Greater Bengaluru (3,026 cells over 583 candidate stormy days).
- Target Variable: flood_label (Binary: 1 = flooded/waterlogged, 0 = assumed dry).
- Overall Class Imbalance: Flood rate is 0.0099% (~1 in 10,000 rows). Exactly 175 positive cell-days exist in the full dataset.
- Validation File: delivery/flood_replay_E2022_09.parquet (72,624 rows covering 24 contiguous days around the catastrophic September 2022 floods).
- All 17 automated acceptance criteria from your spec (Appendix B) have passed with zero failures.

2. Inventory of Delivered Data Files

You have been delivered a clean, ready-to-use directory structure in delivery/:

File Name	Format	Size	Rows	Primary ML Purpose
flood_dataset.parquet	Parquet	18 MB	1,764,158	Primary training & evaluation table for XGBoost.
flood_dataset_event_windows.csv	CSV	111 MB	308,652	All cells on the 102 event window days (-14 to +7 days around events).
flood_replay_E2022_09.parquet	Parquet	921 KB	72,624	Held-out test replay file for September 2022 (24 contiguous days).
cells_static.csv	CSV	268 KB	3,026	Static terrain features and ward lookups for all cells.

File Name	Format	Size	Rows	Primary ML Purpose
rainfall_daily.csv	CSV	1.1 MB	36,112	Continuous daily rainfall (976 days × 37 pixels) for LSTM recurrent training.

Table 1: Description of files provided in the delivery/ directory.

3. Master Feature Schema & Usage Rules

The primary dataset (flood_dataset.parquet) contains 42 columns. For your convenience, the columns are categorized into Features (to feed into the model), Group/Split Keys (for CV only), Labels, and Traceability Audit columns.

3.1 Dynamic Rainfall Features (Feed into Model)

- rain_1d_mm: Rainfall on day d (mm).
- rain_3d_mm: Rolling 3-day sum ($\text{rain}(d-2) + \text{rain}(d-1) + \text{rain}(d)$).
- rain_7d_mm, rain_14d_mm, rain_30d_mm: Rolling cumulative precipitation (mm).
- rain_lag1_mm ... rain_lag14_mm: Daily rainfall on day d-k for k in 1..14 (mm).

Rule: All rolling sums and lags were precomputed from continuous daily time series before row selection. Every row satisfies the invariant: $\text{rain_3d_mm} == \text{rain_1d_mm} + \text{rain_lag1_mm} + \text{rain_lag2_mm}$ ($|\text{diff}| < 0.02$).

3.2 Static Terrain, Hydrologic & Urban Features (Feed into Model)

- elev_m: Mean elevation from raw Copernicus GLO-30 DSM (range: 767.1 to 951.7 metres).
- slope_deg: Mean terrain slope derived from DEM via Horn's method (range: 0.0 to 12.3 degrees).
- twi: Topographic Wetness Index = $\ln(\text{flow_acc_m2} / \tan(\text{slope_rad}))$ (range: 4.8 to 13.1).
- hand_m: Height Above Nearest Drainage to hydrologically conditioned stream channels (range: 0.0 to 58.5 metres).
- flow_acc: Maximum upstream contributing catchment area in km^2 (range: 0.0 to 76.5 km^2).
- imperv_frac: Built-up urban fraction from ESA WorldCover 2021 Class 50 (range: 0.00 to 0.99).
- dist_lake_m: Euclidean distance to nearest OpenStreetMap waterbody polygon (metres).
- dist_drain_m: Euclidean distance to nearest OSM waterway / rajakaluve (metres).
- dist_road_m: Euclidean distance to nearest OSM major road network (motorway/trunk/primary/secondary).
- lakes_within_1km: Count of dissolved OSM water polygons intersecting a 1,000 m centroid radius (integer: 0 to 11).

Rule: Zero missing values exist in static features. Static values are strictly invariant across time for each cell.

3.3 Splitting & Grouping Keys (NEVER Feed as Features)

- cell_id: Integer (0 to 3,025). Do NOT use as a feature (leads to spatial memorization).
- date: ISO string (YYYY-MM-DD). Do NOT use as a feature (temporal memorization).
- lat, lon: WGS84 geographic coordinates. Optional for spatial coordinates, but prefer physical terrain features.
- ward: Municipal ward ID (1 to 198). Use STRICTLY for GroupKFold spatial cross-validation.
- event_id: Code e.g. 'E2022_09'. Use STRICTLY for Leave-One-Event-Out cross-validation.

3.4 Labels & Audit Columns

- flood_label: Binary target (1 = documented flood/waterlogging, 0 = assumed dry).
- label_source: Category ('hotspot_pdf', 'news', or 'none').
- label_confidence: Grade ('high', 'medium', or 'low').
- day_sampling_weight: Float (uniformly 1.0; no downsampling of candidate days was performed).
- audit_is_buffer_cell: Binary flag (0 = report-mapped centre cell, 1 = topological 3×3 neighbour).
- audit_geocode_precision: String ('landuse_footprint', 'polygon_centroid', 'neighbourhood_centroid').

4. Critical Physical Nuances: What Your Model Needs to Know

Your model will fail or learn distorted correlations if you do not account for these three physical realities:

1. CHIRPS Peak Attenuation vs. Ground Cloudbursts:

CHIRPS provides satellite infrared precipitation at ~5.3 km resolution (0.05°). In Bengaluru, urban cloudbursts are hyper-localized convective storms. On the night of 4–5 September 2022, an official IMD city gauge recorded 131.6 mm in 12 hours. In CHIRPS, this appears as 23.1 mm (Pixel 12) and 30.8 mm (Pixel 20) because the rainfall is spatially averaged across 25 km². **CRITICAL MODELING RULE:** Do NOT set hard classification rules expecting >100 mm rainfall. Your model must rely on cumulative 3-day (rain_3d_mm) and 7-day (rain_7d_mm) features combined with low HAND (hand_m < 5 m) and high TWI.

2. The UTC Day Definition & Lag 1 Alignment:

CHIRPS daily precipitation is measured on the UTC calendar day (00:00 to 23:59 UTC), which corresponds to 05:30 IST on day d to 05:30 IST on day d+1. The catastrophic storm on 4 September 2022 peaked around 21:00–02:00 IST (15:30–20:30 UTC), registering inside UTC day 4 September. The catastrophic flooding was documented on the morning of 5 September. **CRITICAL MODELING RULE:** On day 5, the primary flood signal is captured by rain_lag1_mm (the day 4 storm) and rain_3d_mm. Always ensure your feature set includes rain_lag1_mm and rain_3d_mm.

3. Severe Ground-Truth Reporting Asymmetry (False Negatives):

Positive labels exist only where citizens or technical agencies documented flooding (tech corridors like ORR, Sarjapur, Whitefield, Yelahanka, and major arterial underpasses). Unreported valley slums that flooded are labeled 0 ('assumed dry'). **CRITICAL MODELING RULE:** Treat flood_label = 0 as

'unobserved/unreported', not guaranteed dry ground. Evaluate models using Precision-Recall AUC (PR-AUC) and Critical Success Index (CSI), NOT raw accuracy.

5. Effective Sample Size & Recommended Splitting

The ground-truth catalog contains 5 confirmed flood events across 4 years. Here is the exact positive label breakdown:

Event ID	Date Range	Total Positives	Centre-Only (Seed)	Buffer Neighbours (3×3)
E2017_08	2017-08-15 (1 day)	9	1 (Cell 897, Koramangala)	8
E2021_11	2021-11-21 (1 day)	9	1 (Cell 2882, Yelahanka)	8
E2022_05	2022-05-05 (1 day)	7	1 (Cell 555, Rainbow Drive)	6 (Boundary lattice)
E2022_09	2022-09-05 to 09-07 (3 days)	141	18 (6 hotspots × 3 days)	123
E2023_05	2023-05-21 (1 day)	9	1 (Cell 1478, KR Circle)	8
TOTAL	5 Historical Events	175	22 Centre Positives	153 Buffer Positives

Table 2: Breakdown of positive training labels by event and spatial precision.

THE LEAVE-ONE-EVENT-OUT (LOEO) WARNING:

If you hold out event E2022_09 (September 2022) as your validation/test set:

- Test Set: 141 positive cell-days across 3 days.
- Training Set: ONLY 34 positive cell-days remaining across 4 events (only 4 centre-cells-only positives)!

RECOMMENDED MODELING STRATEGY:

1. For XGBoost, set `scale_pos_weight = (len(train_neg) / len(train_pos)) ≈ 10,000`, or train with sample weights:

```
sample_weight = np.where(df.flood_label == 1, 100.0, 1.0)
```

2. If you want high-precision models, you can optionally filter training to centre cells only (`audit_is_buffer_cell == 0`), or downweight buffer cells (sample weight: centre = 1.0, buffer = 0.4, negative = 1.0).

3. Evaluate spatial generalization using GroupKFold(n_splits=5) on ward to ensure the model does not overfit to specific locations.

6. Ready-to-Run Python Training Script (Copy-Pasteable)

Here is a complete, leakage-free training baseline you can immediately execute in Python with XGBoost:

```
import pandas as pd, numpy as np, xgboost as xgb
from sklearn.metrics import precision_recall_curve, auc, classification_report

# 1. Load the dataset
df = pd.read_parquet('delivery/flood_dataset.parquet')

# 2. Define explicit feature list (Zero leakage!)
FEATURES = [
    'rain_1d_mm', 'rain_3d_mm', 'rain_7d_mm', 'rain_14d_mm', 'rain_30d_mm',
    'rain_lag1_mm', 'rain_lag2_mm', 'rain_lag3_mm',
    'elev_m', 'slope_deg', 'twi', 'hand_m', 'flow_acc', 'imperv_frac',
    'dist_lake_m', 'dist_drain_m', 'dist_road_m', 'lakes_within_1km'
]

# 3. Split by Event: Hold out E2022_09 (September 2022 catastrophic floods)
train_df = df[df['event_id'] != 'E2022_09'].copy()
test_df = df[df['event_id'] == 'E2022_09'].copy()

# 4. Calculate class weighting for extreme imbalance
n_pos = (train_df['flood_label'] == 1).sum()
n_neg = (train_df['flood_label'] == 0).sum()
scale_weight = n_neg / max(n_pos, 1)

# Sample weights: trust centre cells fully, discount buffer cells
train_weights = np.where(
    train_df['flood_label'] == 1,
    np.where(train_df['audit_is_buffer_cell'] == 0, 1.0, 0.5),
    1.0
)

# 5. Train XGBoost Classifier
model = xgb.XGBClassifier(
    n_estimators=300,
    max_depth=4,
    learning_rate=0.03,
    subsample=0.8,
    colsample_bytree=0.8,
    scale_pos_weight=scale_weight,
    eval_metric='aucpr',
    random_state=42
)
model.fit(train_df[FEATURES], train_df['flood_label'], sample_weight=train_weights)

# 6. Evaluate on Held-out Event
test_preds = model.predict_proba(test_df[FEATURES])[:, 1]
p, r, _ = precision_recall_curve(test_df['flood_label'], test_preds)
```

```
pr_auc = auc(r, p)
print(f'Test Event PR-AUC: {pr_auc:.4f}')
```

7. Master System Prompt to Feed into Your Coding AI

If you are using Claude, Cursor, Copilot, or ChatGPT to build your models, copy and paste the box below as your System Prompt. It provides complete context so your AI will not make beginner errors:

```
=== SYSTEM PROMPT FOR MODELING AI ===
```

```
You are an expert machine learning engineer assisting Member 2 in building an XGBoost and LSTM predictive flood model for Bengaluru, India (Community Project 1BCP308, Team N2M). Member 1 has delivered the verified training dataset.
```

```
DATASET SPECIFICATIONS:
```

1. Master file: 'delivery/flood_dataset.parquet' (1,764,158 rows, 42 columns, 3,026 spatial cells, 583 candidate stormy days).
 2. Ground truth target: 'flood_label' (0 or 1). Extreme class imbalance: 175 positives out of 1.76M rows (~0.0099%).
 3. Physical features to use: rain_1d_mm, rain_3d_mm, rain_7d_mm, rain_14d_mm, rain_30d_mm, rain_lag1_mm..14_mm, elev_m, slope_deg, twi, hand_m, flow_acc, imperv_frac, dist_lake_m, dist_drain_m, dist_road_m, lakes_within_1km.
 4. DO NOT use date, cell_id, ward, or event_id as features (use ward for spatial GroupKFold, event_id for event cross-validation).
 5. Physical Nuance: CHIRPS daily rain (~5.3 km) attenuates extreme single-day cloudbursts (a 131 mm storm registers as ~25-30 mm). Rely heavily on cumulative sums (rain_3d_mm, rain_7d_mm) and lag 1.
 6. Validation: Hold out event 'E2022_09' for test replay using 'delivery/flood_replay_E2022_09.parquet' (72,624 rows).
 7. Class weighting: Use scale_pos_weight or weighted cross-entropy. Do not evaluate with raw accuracy; use PR-AUC and Critical Success Index (CSI).
- ```
=== END SYSTEM PROMPT ===
```